

Customer Segmentation & Churn Prevention Roadmap

Actionable business strategies derived from K-Means clustering (k=3), Agglomerative hierarchical models, and segment-specific tuned Logistic Regression classifiers.

DATASET SIZE 500 Account Records Analysed	OVERALL CHURN RATE 10.6% 53 Observed Churn Cases	MEAN MONTHLY CHARGES \$11 3.64 Range: \$20.00 – \$1 99.00	AVG. CUSTOMER TENURE 36.5 mo Approx. 3 Years Lifetime
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1 Customer Segment Profiles (K-Means k=3)

Segment 0: Established Loyalty 114 Customers (22.8%) Mature customer base anchored by multi-year agreements. Characterised by longer tenure, lower baseline churn propensity, and credit card / automated billing preference. Primary Driver: Tenure (-4.02) & 2-Yr Contract (-2.47)	Segment 1 : High-Volume Core 249 Customers (49.8%) The core volume segment. Driven by monthly plans and electronic payments. Exhibits acute demographic sensitivity; high responsiveness to contract lock-in incentives. Primary Driver: Senior Citizen (-4.55) & Tenure (-3.1 8)	Segment 2: High-Velocity Risk 1 37 Customers (27.4%) High average monthly charges paired with volatile tenure. The most vulnerable tier where pricing friction and month-to-month contracts trigger severe customer exits. Primary Driver: Tenure (-7.99) & Senior Citizen (+4.83)
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2 Segment-Specific Model Diagnostics (Tuned Classifiers)

Core Analytical Takeaway One-size-fits-all churn prediction fails to capture diverging churn triggers. Segment 1 achieves high F1 reliability (0.875), while Segment 0's perfect precision (1.00) enables zero-waste targeted interventions. Segment 2 requires aggressive tenure-building programs to mitigate early account degradation.
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3 Segment Behavioral Drivers (Feature Coefficients)

Feature Name	Segment 0 (Loyal)	Segment 1 (Core Volume)	Segment 2 (High-Risk)	Strategic Significance
Tenure	-4.0239	-3.1760	-7.9866	Strongest universal retention factor across all tiers.
Senior Citizen	+1 .8249	-4.5451	+4.831 6	Severe attrition risk in Seg 2; strong loyalty in Seg 1 .
Two-Year Contract	-2.4702	0.0000 (Sparsity)	-2.6900	Potent insulation against competitive poaching.
Month-to-Month Contract	+1 .6727	+1 .8626	0.0000 (Sparsity)	Primary catalyst for short-term exit decisions.
Monthly Charges	Negligible	+0.4593	+2.0695	Significant price sensitivity triggering churn in Segment 2.
Paperless Billing (Yes/No)	-1 .561 9	-0.8258	-0.3286	Standard paper invoicing correlates with lower churn rates.

4 Targeted Strategic Business Recommendations

Segment 0: Value Retention & Advocacy **Tier A Priority**

- 1 Multi-Year Contract Renewal Bundles:** Proactively target customers 60 days before contract expiry with tenure-loyalty rebates to preserve the -2.47 protective coefficient.
- 2 Automated Payment Protection:** Incentivize ACH/Credit Card continuity with annual billing discounts; payment friction is a silent exit vector.

Segment 1: Migration to Long-Term Commitments

Tier A Priority

- 1 Contract Escalation Pathways:** Convert Month-to-Month users (+1.86 churn coefficient) into 1-Year terms via waived activation fees or feature upgrades.
- 2 Senior Care Programs:** Capitalize on the strong negative churn coefficient (-4.55) through dedicated senior support desks and tailored digital onboarding.

Segment 2: Price De-escalation & Churn **Urgent Focus**

- 1 High-Charge Optimization:** High monthly spend (+2.07) drives exits. Introduce modular pricing tiers to right-size bills before billing shock triggers departure.
- 2 First-Year Tenure Milestone Shields:** Tenure is the ultimate defense (-7.99). Deploy customer-success check-ins at months 3, 6, and 12 to cross the critical tenure threshold.

Enterprise Strategy: Early Warning Pipeline

Systemic Rollout

- 1 Segment-Specific Scoring APIs:** Ingest live CRM telemetry into the three tuned Logistic Regression models to generate real-time churn probability scores.
- 2 Churn Interception Playbooks:** Trigger automated retention offers when churn probability crosses 0.60, matching incentives to segment-specific coefficient profiles.

5 90-Day Implementation & Financial Impact Roadmap

Phase	Timeline	Key Milestones & Deliverables	Projected Outcome
Phase 1: Triage	Days 1 – 30	Deploy Segment 2 pricing reviews; contact top 20% highest-charge accounts nearing exit probability thresholds.	Immediate 15–20% churn reduction in High-Risk tier.
Phase 2: Migration	Days 31 – 60	Launch Month-to-Month conversion campaign across Segment 1; offer 12-month lock-in incentives.	Shift 25% of monthly cohort into stable contract tiers.
Phase 3: Automation	Days 61 – 90	Automate churn scoring models in production CRM; integrate renewal triggers into billing workflows.	Sustained drop of overall churn from 10.6% to ≤7.0%.